Project 2: Magnitude Spectrum Inversion

This project demonstrates the reconstruction of audio signals from magnitude-only spectrum data, without phase information. It uses the strategies of the Real-Time Iterative Spectrogram Inversion (RTISI) and added look-ahead method introduced by Zhu, Beauregard and Wyse in 2007. These algorithms build on the previous state-of-the-art from Griffin and Lim in 1984, and provide notable practical improvements in performance. Replication of these steps, along with analysis and evaluation, is described sequentially in this report.

Zhu, Beauregard & Wyse Paper

The stated goal of Zhu, Beauregard, and Wyse, is to accomplish the same output as Griffin and Lim, namely reconstructing a sound signal from spectrogram data by determining the best phase values to apply to the data. Zhu et al. want to overcome some of the limitations of the solution from Griffin and Lim, to be usable near real-time, and also flexible for various computing and performance environments. They present a combination of strategies to meet their list of requirements.

1. Real-Time Iterative Spectrogram Inversion (RTISI) Algorithm

The simple RTISI process uses overlapping frames to iteratively refine guesses of the best value for the current frame's phase. Each frame is fitted with a Hamming window, which uses the formula:

(Available from many sources, including SciPy documentation: https://docs.scipy.org/doc/scipy/reference/generated/scipy.signal.windows.hamming.html)

$$w(n) = 0.54 - 0.46\cos\left(\frac{2\pi n}{M-1}\right) \qquad 0 \le n \le M-1$$

With step size set at 1/4 frame width, the windowing function allows the overlapping signals to be added together to approximate the corresponding segment of the original sound file. RTISI processes each frame by iterating through and improving guesses for the phase value, added to the overlapping part of the already-processed signal to test and refine the guesses. Each comparison requires an inverse transform into time domain to test the phase value, and a transform to Fourier domain again to calculate the improved phase estimate and prepare for the next loop. Zhu et al. initialize the first frame with a phase guess of zero, which follows Griffin and Lim's method. However, by initiating each subsequent frame with the phase value from its preceding and overlapping neighbor, Zhu et al. are able to get high-quality results with far fewer iterations. The computational costs of the whole process are characterized by the two transformations performed per iteration.

2. RTISI with Look-Ahead (RTISI-LA) Algorithm

RTISI provides a partial solution to the improvements Zhu et al. desired over Griffin and Lim, but it does not meet their flexibility and quality requirements. The simple RTISI process is limited by its asymmetry; the information available when guessing the phase of a frame is weighted towards the beginning of the frame, and no comparison is

available at all for the final 25% of the frame. Zhu et al. improve their initial algorithm by adding a look-ahead capability. This means introducing a buffer to delay committing to the phase value for a frame until a set number of frames beyond it have been processed. This means for each additional look-ahead frame, the preceding uncommitted frames go through another loop of iterations. While this gives more opportunity to refine the phase values, it also multiplies the number of transformations computed and delays the availability of results.

3. Implementation Strategies

Zhu et al. use the signal-to-error ratio (SER) function from Griffin and Lim to evaluate their outputs for different hyperparameter choices:

$$10\log\frac{\sum_{m=-\infty}^{\infty}\frac{1}{2\pi}\int_{\varpi=-\pi}^{\pi}|X(mS,\varpi)|^2 d\varpi}{\sum_{m=-\infty}^{\infty}\frac{1}{2\pi}\int_{\varpi=-\pi}^{\pi}[|X(mS,\varpi)|-|X'(mS,\varpi)|]^2 d\varpi}$$

They report results for RTISI-LA on speech and music samples, all using a frame size of 23.2 ms. They report increasing SER as look-ahead frame count is raised, and similarly, improving SER as iterations increase. These demonstrate a common theme, the trade-off between performance and quality. The more time and computation applied, the better results can be produced, but these decisions need to be made with the time, computing, and quality requirements of an application in balance. All SER values are reported for speech and music separately in the Zhu, Beauregard, and Wyse paper.

Window Parameters

The primary hyperparameter is the window or frame size, which combines with the step size to determine the total number of frames that will be created from the original audio, and how much they will overlap. In my implementation, as Zhu et al. emphasized, I always select a step size 1/4 the length of the frame size. I wrote the code to allow these values to be set in milliseconds (ms) for each trial. I tested a range of frame sizes, always using a Hamming window on each overlapping frame, while varying other parameter settings.

Scaling Strategy

The input audio is provided in mp3 format for both speech and music samples. To process them and transform into magnitude spectrum data in Fourier domain, I first load them as 16-bit integers and convert to float32 to conform with the calculations that will follow. The transformed signals are processed with their original amplitude scale. However, when it comes time to output the final results of phase reconstruction in time domain, the signals are put through additional processing. They are normalized with room added above max amplitude to prevent clipping, then scaled back to 16-bit audio and saved as mp3 files, the same format as the original audio.

Performance/Speed Comparison

In addition to the design factors discussed above, the set number of iterations is a primary hyperparameter for both RTISI and RTISI-LA. Because of the buffer design, the look-ahead algorithm performs (iteration value * look-ahead frame count) iterations for each frame, excluding those at the beginning and end of the sound file that have fewer overlapping neighbors. Therefore, RTISI-LA achieves comparable quality with a lower iteration setting, but also takes longer to run. This is a prime example of the trade-off between performance cost and speed, as well as performance cost and output quality. Ideally, depending on the requirements for a particular use case, these would be set to achieve the most efficient and effective results for the characteristics of sound files that will be processed. If varying types of sound files are included, it would make sense to provide flexibility in the selected parameters, along with guidance for the user about which settings are expected to meet their needs for the expected range of audio characteristics.

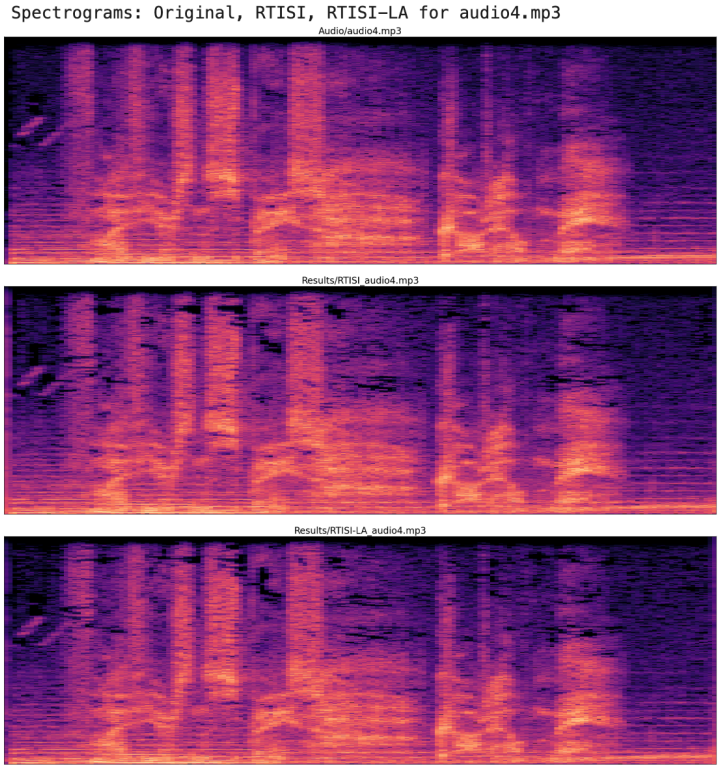
Reconstruction Quality Evaluation

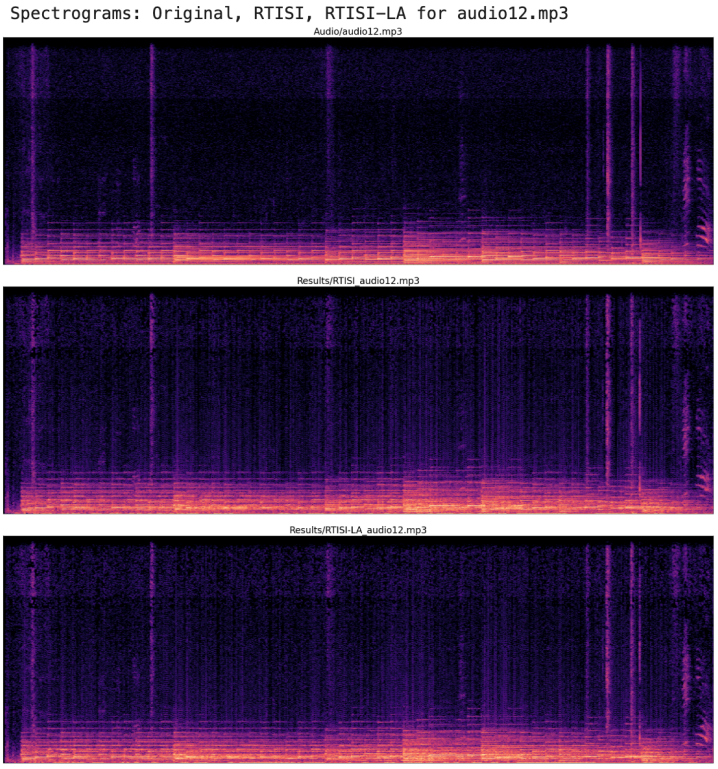
To compare and evaluate results, I implemented SER following the same formula as used by Zhu et al. (provided above on page 2). I also attempted adding Perceptual Evaluation of Speech Quality (PESQ) from https://pypi.org/project/pesq/, in addition to SER. Ultimately I did not use it because the library requires specific sampling rates (8k or 16k), which would mean resampling for every measurement. I would not expect it to provide significant functionality to the code in this context. This is because from what I read about this measurement, it is valid for relatively clean recordings of speech. However, the speech samples provided for this project generally have background music or other sounds, which are not ideal for PESQ validity.

The SER tables reported by Zhu et al. demonstrate increasing quality but diminishing speed for higher values of both look-ahead frames and iterations. However, they do not report any results for RTISI without look-ahead, and they do not compare different frame sizes for RTISI-LA. I experimented with a variety of hyperparameter selections to learn how they affect different sound files, then I systematically compared a range of frame sizes on the given sound files. Variation in frame size potentially presents another trade-off between cost and quality, because smaller frames might be hypothesized to provide higher-quality results, but the total number of program calculations will double as frame size is cut in half. However, my experiments show that the relationship between frame size and quality is more complex, and there may be different ideal frame sizes not only for speech and music, but even differences between types of music and speech recordings.

As I refined my model functions, I noticed some of the sound files produced many more artifacts than others. In addition, the background sounds accompanying speech samples seemed particularly challenging to reconstruct. After I got the models performing as expected, I compared a variety of options for frame size and iterations for both models, as well as look-ahead frames for RTISI-LA. I found some interesting results. In general, RTISI-LA with 3 look-ahead frames produced higher-quality results

Spectrograms from 40 ms trial: audio4.mp3 (speech with background music) and audio12.mp3 (sparse piano music):





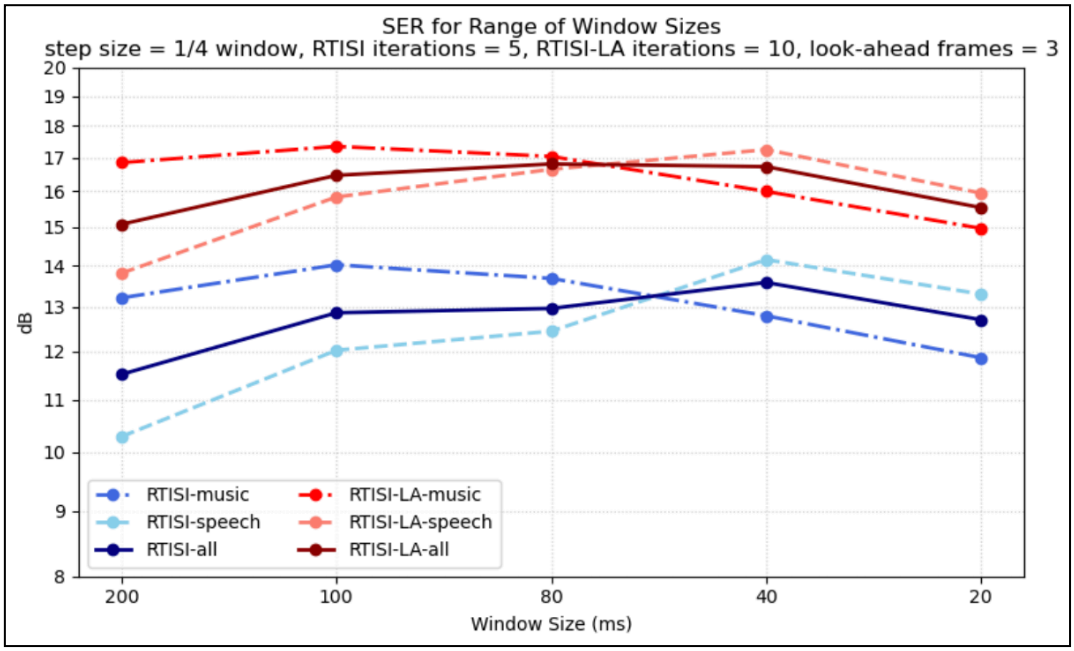
than RTISI, even with iterations set at half. Also, for RTISI-LA with a frame size of 100 ms, I found no change in SER when I increased iterations from 10 to 20.

I decided to test a range of frame sizes while keeping the other parameters steady. I tested five frame sizes from 20 to 200 ms. For RTISI I used 10 iterations, for RTISI-LA I used 5 iterations and 3 look-ahead frames, and I always set the step size to 1/4 of the frame size. In addition to calculating SER for each result, I plotted spectrograms of the reconstructed audio alongside the spectrogram of the original sound file. Most of the groups of spectrograms superficially look the same to me. However, I did note some visually apparent differences in the particularly challenging audio samples. These can be seen above for the 40 ms frame size trial. On the left are the spectrograms for audio4.mp3, a speech file with background music. The original is on top, RTISI in the middle, and RTISI-LA below. There are more areas that look like black voids in both reconstructed examples, which are not present in the original sound file. The spectrograms on the right are from audio12.mp3, a sample of sparse piano music. In these plots, the reconstructions show many more dark purple streaks reaching towards the top of the plot, which are not present in the original.

While looking at the SER results for each individual sound file, it was interesting to observe that the ‘best’ frame size tested varied significantly, even between different music samples, and different speech samples. In fact, the results for audio4.mp3 appear particularly complex, peaking at 100 ms for both RTISI and RTISI-LA, but also

increasing again for the smallest frame sizes. This could be due to the presence of both speech and music in this sample.

I took the SER results from the varying frame sizes, and aggregated the scores into music, speech, and all, for each of the models. These results are plotted below.



Although results for each individual file were unique, there are some clear patterns suggesting hypotheses that could be tested in future work. In all cases, the RTISI-LA results were better than RTISI, as expected. However, for both models, the best results for the music samples were achieved at 100 ms frame size. For speech, the best results came from 40 ms frame size. This suggests parameters should be adjusted based on audio characteristics. Interestingly, it also appears to be less costly computationally to achieve comparable results for music, since larger frame sizes mean fewer total transformations must be computed.

The SER values measured by my program are marginally below those reported by Zhu, Beauregard and Wyse. However, this could be due to differences in sound files used. For example, all of our sample files are given in mp3 format, which may be lower quality than the recordings researchers used. In addition, our speech recordings have a number of background sounds, including music, which were not reported as characteristics of the speech files used by Zhu et al.

Conclusion

Overall, this project provided a challenge and yielded some surprising results. Based on the plotted patterns in SER for a limited selection of frame sizes, I hypothesize that there might be clusters amongst types of music and speech recordings, which would show different ideal frame sizes. I also wonder about the possibilities of implementing hybrid strategies, for example splitting the audio above and below a frequency threshold and using different parameters for each resulting signal.

Sources:

Bojkovic, Z. S., Bakmaz, B. M., & Bakmaz, M. R. (2017). “Hamming window to the digital world.” *Proceedings of the IEEE*, *105*(6), 1185-1190.

Griffin, D., & Lim, J. (1984). “Signal estimation from modified short-time Fourier transform.” *IEEE Transactions on acoustics, speech, and signal processing*, *32*(2), 236-243.

Zhu, X., Beauregard, G. T., & Wyse, L. L. (2007). “Real-time signal estimation from modified short-time Fourier transform magnitude spectra.” *IEEE Transactions on audio, speech, and language processing*, *15*(5), 1645-1653.